

Interleaved streams

Deterministic recovery of padded prefixes



Uniform gap and pruning

Sequential table selection; short returns removed



Heisenberg voltages and spectral sampling

Circuit separation and full regular operator norm control



Geometric realization

Embedded sheets; disjoint domains; finite cycle coordinates



Integral detector and torsion control



Supported Bott homotopies

A primitive torus class in a torsion-free group Even and odd families agree; initial class vanishes



A finite witness gives a finitely generated subgroup

$$h \otimes 1 \neq 0 \text{ in } K_0(BG_{\text{inj}}) \otimes \mathbb{Q}, \quad \mu_{G_{\text{inj}}}^r(h) = 0$$